



Lecture 14

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Mathematical preliminaries for Support Vector Machines

- ▶ Constrained optimization and Lagrange multipliers.
- ▶ Primal and dual problems and how their solutions are related
- ▶ Karash-Kuhn-Tucker conditions.
- ▶ Definition of Kernel Functions
- ▶ Linear Classifiers



We shall work with the following optimization problem:

$$\begin{aligned} &\text{minimize } f(\mathbf{x}) \text{ subject to} \\ &g_i(\mathbf{x}) \leq 0 \quad \forall i \in [m] \\ &h_j(\mathbf{x}) = 0 \quad \forall j \in [p] \end{aligned}$$



The Lagrangian associated with this optimization problem is



$$\text{minimize } f(\mathbf{x}) + \sum_{i=1}^{i=m} \lambda_i g(\mathbf{x}) + \sum_{j=1}^{j=p} \nu_j h_j(\mathbf{x})$$

- ▶ The λ_i 's and h_j 's are called Lagrange multipliers.



- ▶ We now consider the case of a quadratic objective function subject to affine constraints:

$$\min_{\mathbf{x} \in \mathbb{R}^d} \frac{1}{2} \mathbf{x}^T \mathbf{Q} \mathbf{x} + \mathbf{c}^T \mathbf{x} \text{ subject to}$$
$$\mathbf{A} \mathbf{x} \leq \mathbf{b}$$

- ▶ Here $\mathbf{A} \in \mathbb{R}^{m \times d}$, $\mathbf{b} \in \mathbb{R}^m$, $\mathbf{c} \in \mathbb{R}^d$



- ▶ The Lagrangian $\mathcal{L}(\mathbf{x}, \boldsymbol{\lambda})$ is given by $\frac{1}{2}\mathbf{x}^T \mathbf{Q}\mathbf{x} + \mathbf{c}^T \mathbf{x} + \boldsymbol{\lambda}^T (\mathbf{A}\mathbf{x} - \mathbf{b})$.
- ▶ Rearranging the above we have $\mathcal{L}(\mathbf{x}, \boldsymbol{\lambda}) = \frac{1}{2}\mathbf{x}^T \mathbf{Q}\mathbf{x} + (\mathbf{c} + \mathbf{A}^T \boldsymbol{\lambda})^T \mathbf{x} - \boldsymbol{\lambda}^T \mathbf{b}$
- ▶ Taking the derivative of $\mathcal{L}(\mathbf{x}, \boldsymbol{\lambda})$ and setting it equal to zero gives $\mathbf{Q}\mathbf{x} + (\mathbf{c} + \mathbf{A}^T \boldsymbol{\lambda}) = 0$.



- ▶ If we take $\mathbf{Q}$ to be invertible, we have $\mathbf{x} = \mathbf{Q}^{-1}(\mathbf{c} + \mathbf{A}^T \boldsymbol{\lambda})$.
- ▶ Plugging this value of $\mathbf{x}$ into $\mathcal{L}(\mathbf{x}, \boldsymbol{\lambda})$ gives us
$$\mathcal{D}(\boldsymbol{\lambda}) = -\frac{1}{2}(\mathbf{c} + \mathbf{A}^T \boldsymbol{\lambda})\mathbf{Q}^{-1}(\mathbf{c} + \mathbf{A}^T \boldsymbol{\lambda}) - \boldsymbol{\lambda}^T \mathbf{b}.$$
- ▶ This gives us the dual optimization problem:
$$\max_{\boldsymbol{\lambda} \in \mathbb{R}^m} -\frac{1}{2}(\mathbf{c} + \mathbf{A}^T \boldsymbol{\lambda})\mathbf{Q}^{-1}(\mathbf{c} + \mathbf{A}^T \boldsymbol{\lambda}) - \boldsymbol{\lambda}^T \mathbf{b} \text{ subject to } \boldsymbol{\lambda} \geq \mathbf{0}.$$



- ▶ Weak duality establishes an inequality connecting primal and dual problems
- ▶ Weak duality condition states that the optimal solution of the primal problem is greater than or equal to that of the dual problem.
- ▶ In the Quadratic Optimization problem discussed previously , weak duality exists



- ▶ Strong duality condition states that the optimal solution of the primal problem is equal to that of the dual problem
- ▶ One can solve the dual problem to get the same solution as solving the primal problem.
- ▶ In some optimization problems, solving the dual problem may be easier.
- ▶ Question: When does strong duality hold?



- ▶ For a primal optimization problem we say that it obeys Slater's condition if
 1. the objective function f is convex, the constraints g_i are all convex, the constraint functions h_i are all linear
 2. there exists a point $\bar{x}$ in the interior of the region, i.e $g_i(\bar{x}) < 0$ for all $i \in [m]$ and $h_j(\bar{x}) = 0$ for all $j \in [p]$.
- ▶ Suppose Slater's condition holds then we have strong duality.

Example :

$$\begin{aligned} \min \quad & x^4 + y^4 \\ \text{st} \quad & x^2 + y^2 - 3 \leq 0 \end{aligned}$$

$$\min f(\mathbf{x}) \quad \text{st} \quad g_i(\mathbf{x}) \leq 0 \quad \forall i \in [m], \quad h_j(\mathbf{x}) = 0 \quad \forall j \in [p]$$

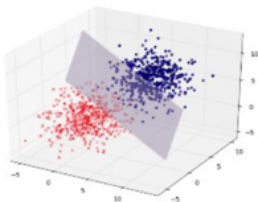
We say that $\mathbf{x}^*$ and $(\lambda^*, \nu^*) \in \mathbb{R}^m \times \mathbb{R}^p$ respect the Karash-Kuhn-Tucker conditions if:

1. $g_i(\mathbf{x}^*) \leq 0 \quad \forall i \in [m], \quad h_i(\mathbf{x}^*) = 0 \quad \forall i \in [p]$.
2. $\lambda_i^* \geq 0 \quad \forall i \in [m]$.
3. $\lambda_i^* g_i(\mathbf{x}^*) = 0 \quad \forall i \in [m]$.
4. $\nabla f(\mathbf{x}^*) + \sum_{i=1}^m \lambda_i^* \nabla g_i(\mathbf{x}^*) + \sum_{i=1}^p \nu_i^* \nabla h_i(\mathbf{x}^*) = 0$.

If strong duality holds then any primal optimal solution $\mathbf{x}^*$ and dual optimal solution (λ^*, ν^*) satisfy the KKT conditions.

$$\mathbf{w}^T \mathbf{x} = 0$$

Hyperplane



$$y = ax + b$$

Line





- ▶ Consider line $w^T x + b = 0$. Let x_a and x_b lie on this line. So $w^T x_a + b = 0$ and $w^T x_b + b = 0$.
- ▶ This means $w^T (x_a - x_b) = 0$. $x_a - x_b$ lies on the line and is directed from x_b to x_a .
- ▶ Hence w is orthogonal to $x_a - x_b$ and in turn, to the line.



- ▶ Kernel is a continuous function $K(x, y)$
- ▶ Kernel takes two arguments x and y
- ▶ x and y could be real numbers, functions, vectors, etc
- ▶ $K(x, y)$ maps x and y to a real value
- ▶ Kernel value is independent of the order of the arguments, i.e.,

$$K(x, y) = K(y, x)$$



- ▶ A kernel function is some function that corresponds to an inner product in some expanded feature space.

$$K(x_i, x_j) = \phi(x_i)^T \phi(x_j)$$

- ▶ Linear classifier relies on dot product between vectors $x_i^T x_j$
- ▶ If every data point is mapped into high-dimensional space via some transformation $\phi : x \rightarrow \phi(x)$, the dot product becomes:
 $K(x_i, x_j) = \phi(x_i)^T \phi(x_j)$



- ▶ For some functions $K(x_i, x_j)$ checking $K(x_i, x_j) = \phi(x_i)^T \phi(x_j)$ is difficult.
- ▶ Mercer's theorem: Every positive-semidefinite symmetric function is a kernel function.



1) We can *construct kernels from scratch*:

- For any $\varphi : \mathcal{X} \rightarrow \mathbb{R}^m$, $k(x, x') = \langle \varphi(x), \varphi(x') \rangle_{\mathbb{R}^m}$ is a kernel.
- If $d : \mathcal{X} \times \mathcal{X} \rightarrow \mathbb{R}$ is a *distance function*, i.e.
 - $d(x, x') \geq 0$ for all $x, x' \in \mathcal{X}$,
 - $d(x, x') = 0$ only for $x = x'$,
 - $d(x, x') = d(x', x)$ for all $x, x' \in \mathcal{X}$,
 - $d(x, x') \leq d(x, x'') + d(x'', x')$ for all $x, x', x'' \in \mathcal{X}$,

then $k(x, x') := \exp(-d(x, x'))$ is a kernel.



2) We can *construct kernels from other kernels*:

- if k is a kernel and $\alpha > 0$, then αk and $k + \alpha$ are kernels.
- if k_1, k_2 are kernels, then $k_1 + k_2$ and $k_1 \cdot k_2$ are kernels.

Examples of Kernels

- ▶ Linear: $K(x_i, x_j) = x_i^T x_j$
- ▶ Polynomial of power p : $K(x_i, x_j) = (1 + x_i^T x_j)^p$
- ▶ Sigmoid: $K(x_i, x_j) = \tanh(\beta_0 x_i^T x_j + \beta_1)$

Classification Problem in Machine Learning



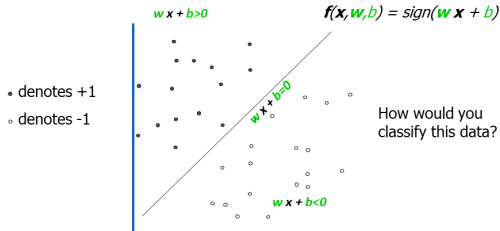
- ▶ Classification of data into different classes is one of the primary problems in machine learning
- ▶ Binary classification involves classifying data into exactly 2 classes
- ▶ There exists different algorithms for binary classification
- ▶ We will discuss on a model called Support Vector Machine.
- ▶ SVM is an instance of a linear classifier model for binary classification

Linear Classifier for Binary Classification



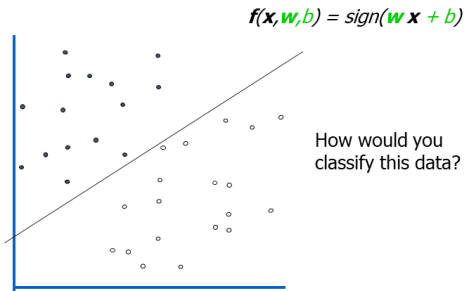
SVM is a linear classifier

Linear Classifiers



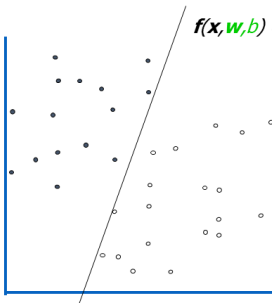
Linear Classifiers

- denotes +1
- denotes -1



Linear Classifiers

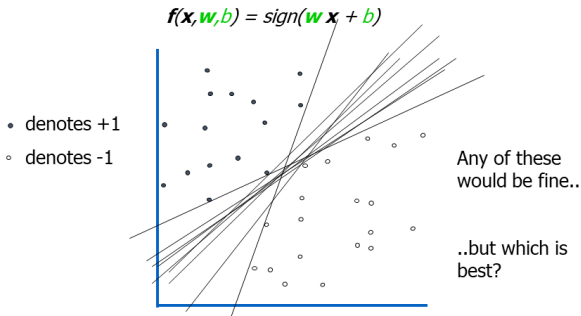
- denotes +1
- denotes -1



$$f(\mathbf{x}, \mathbf{w}, b) = \text{sign}(\mathbf{w} \mathbf{x} + b)$$

How would you classify this data?

Linear Classifiers



Linear Classifiers

- denotes +1
- denotes -1

